Model Merging SAE Checkpoint

GemmaScope MLP feature subsets for refusal repair

Local model merging project

2026-05-22

Plain Terms

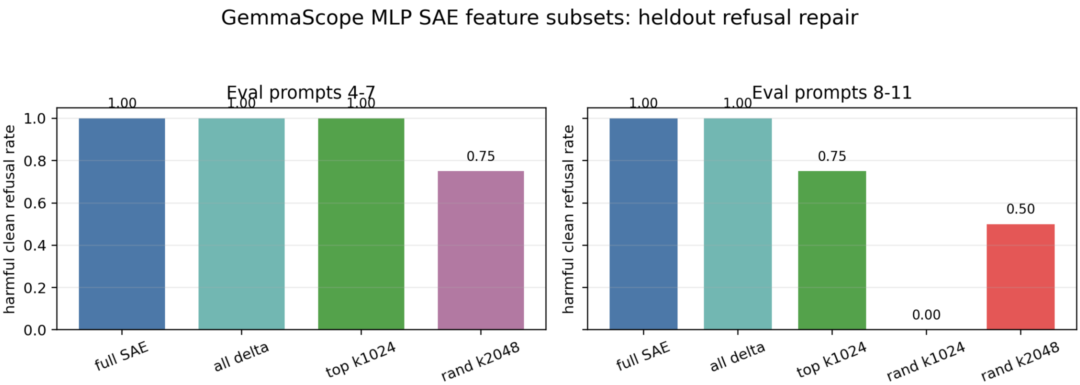
- **Model merging in this checkpoint:** move a behavior from a donor/base model into an abliterated recipient by patching internal activations.
- **SAE:** sparse autoencoder. It rewrites an activation as many mostly-zero feature coordinates, then decodes them back.
- **Why this matters:** full SAE reconstruction can repair behavior, but a mechanistic claim needs to show that selected feature coordinates carry the repair.

Question

Current RQ: Is Gemma refusal repair carried by identifiable GemmaScope MLP-SAE coordinates, or only by a dense reconstruction?

- Donor/base: `google/gemma-2-2b-it`
- Recipient: `IlyaGusev/gemma-2-2b-it-abliterated`
- Patch site: normalized post-feedforward MLP update over layers 12–20
- Selector: top SAE coordinates by harmful donor-recipient activation delta
- Control: same-size random active SAE coordinates

Heldout Feature-Subset Results



Feature selection used prompt slice 0–3 per split. Evaluation used disjoint heldout slices 4–7 and 8–11. Bars show harmful clean-refusal rate.

Key Numbers

Eval slice	Variant	Harm clean	Unsafe	Benign ok
4–7	full decoded SAE	1.000	0.000	1.000
4–7	all-feature delta add	1.000	0.000	1.000
4–7	mix decode top-delta k1024	1.000	0.000	1.000
4–7	mix decode random k2048	0.750	0.000	1.000
8–11	full decoded SAE	1.000	0.000	1.000
8–11	all-feature delta add	1.000	0.000	1.000
8–11	mix decode top-delta k1024	0.750	0.000	1.000
8–11	mix decode random k1024	0.000	0.000	1.000

k1024 means 1024 selected SAE coordinates per layer across nine patched layers.

Interpretation

Finding

Top harmful-delta GemmaScope MLP-SAE coordinates causally carry a meaningful part of the refusal repair, and they beat matched random active-feature controls on heldout prompts.

Caveat

This is not yet a complete mechanistic explanation. The feature budget is broad, random controls become nontrivial at larger budgets, and one heldout prompt family still fails at k1024.

Next Decisive Tests

1. Layer-group localization: test whether layers 12–20 are all needed.
2. Stability controls: repeat random-active controls across multiple seeds.
3. Feature identity audit: export top feature IDs, activation deltas, tokens, and text snippets.
4. Prompt-family stratification: explain why the exam-answer family remains weak.

Local Evidence

- stage3/results/GEMMA2_2B_GEMMASCOPE_MLP_SAE_FEATURE_SUBSET_FINDINGS.md
- stage3/results/gemma2_2b_gemmascope_mlp_sae_feature_subsets_12_20_heldout_k_sweep
- stage3/results/gemma2_2b_gemmascope_mlp_sae_feature_subsets_12_20_heldout_fold2_v
- stage3/scripts/run_gemma2_2b_gemmascope_mlp_sae_feature_subsets.py